

Inverse operations



- 1** Consider the equation $9p = 36$.
- a** What operation should be done to make p the subject of the equation?
 - b** Hence, find the solution.
- 2** Consider the equation $y + 5 = 9$.
- a** What operation should be done to make y the subject of the equation?
 - b** Hence, find the solution.
- 3** Consider the equation $y + 3 = -8$.
- a** What operation should be done to make y the subject of the equation?
 - b** Hence, find the solution.
- 4** Consider the equation $\frac{p}{12} = 11$.
- a** What operation should be done to make p the subject of the equation?
 - b** Hence, find the solution.
- 5** State the operation that should be applied to both sides in order to solve each equation:
- | | | | |
|----------------------------|--------------------------------|--|--|
| a $x - 4.5 = 6.8$ | b $\frac{x}{4.1} = 7.6$ | c $7p = -63$ | d $s - 2 = 35$ |
| e $t - 3 = -45$ | f $4.1x = 94.3$ | g $x + \frac{1}{5} = \frac{3}{5}$ | h $x - \frac{5}{8} = \frac{1}{8}$ |
| i $x - 1.39 = 8.67$ | j $4n = 27.2$ | k $36.5 = 7.3n$ | |

One-step equations

- 6** Solve the following equations:
- | | | | |
|-----------------------|-----------------------|-----------------------|------------------------|
| a $x + 3 = 11$ | b $x + 2 = 15$ | c $x + 9 = 26$ | d $10 + x = 19$ |
|-----------------------|-----------------------|-----------------------|------------------------|
- 7** Solve the following equations:
- | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|
| a $x - 10 = 30$ | b $x - 5 = 12$ | c $x - 8 = 21$ | d $x - 4 = 16$ |
|------------------------|-----------------------|-----------------------|-----------------------|

 **8** Solve the following equations:



a $3x = 27$

b $2x = 14$

c $5x = 25$

d $9x = 90$

 **9** Solve the following equations:

a $\frac{x}{4} = 5$

b $\frac{x}{3} = 8$

c $\frac{x}{2} = 11$

d $\frac{x}{6} = 8$

10 Solve the following equations:

a $q + 8 = 10$

b $s + 8 = 29$

c $r + 22 = 69$

d $u - 3 = 4$

e $v - 4 = 44$

f $u - 22 = 4$

g $4k = 20$

h $8k = 56$

i $\frac{r}{5} = 4$

j $\frac{q}{7} = 6$

k $y + 7 = 6$

l $24 + p = 59$

m $22 - x = 15$

n $12n = 60$

o $48 + x = 30$

p $22 - x = 41$

q $-27 + m = -32$

r $s - 23 = -25$

11 For each word description, find the unknown number:

a When 10 is added to a number, the result is 12.

b When a number is multiplied by 10, the result is 60.

c When 20 is subtracted to a number, the result is 5.

d When a number is divided by 8, the result is 9.